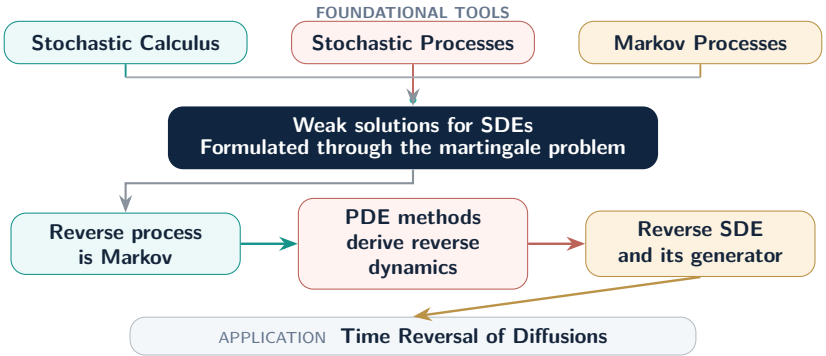
Operator semigroups

Time Reversal of Diffusions

Subtitile

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From foundations to time reversal



Markov Chains

The transition function of a Markov Chain $(X_n)_{n\in\mathbb{N}_0}$ is given by

$$P(x,y) := \mathbb{P}(X_{n+1} = x \mid X_0 = x_0, \ldots, X_n = y) \overset{Markov}{=} \mathbb{P}(X_{n+1} = x \mid X_n = y).$$

Invariant Measures

A measure π on E is called invariant measure wrt. P if

$$\forall x \in E: \quad \pi(x) = \sum_{y\in E} \pi(y)p(y,x).$$

By definition

$$\pi = \pi P^n.$$

So if we initialize the Markov-Chain $(X_n)_{n\in\mathbb{N}_0}$ with transition function P with the invariant measure π:

$$\forall x \in E: \quad \mathbb{P}_\pi(X_n = x) := \sum \pi(y)\mathbb{P}_y(X_n = x) = \sum \pi(y)P^n(y,x) = \pi(x)$$

Intuition: The invariant measure describes the equilibrium distribution of the Markov-Chain.

Markov Chains

Reversibility

A Markov Chain is reversible if going forward or backward in time in equilibrium is the same:

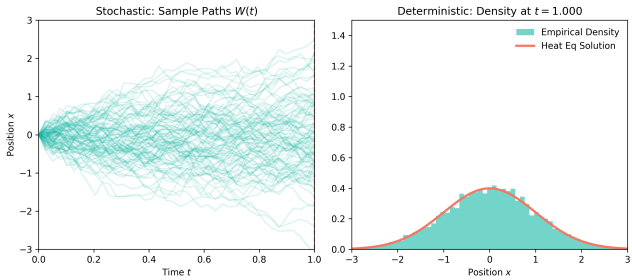
$$\forall x, y \in E: \quad \pi(x)P(x,y) = \pi(y)p(y,x).$$

Denote by $(X_n)_{n=1,\dots,N}$ a Markov-Chain with transition Matrix P up to time $N : \Omega \to \mathbb{N} \cap \{\infty\}$. The Backward Markov-Chain $(\tilde{X}_n := X_{N-n})_{n=1,\dots,N}$ has the transition Matrix

$$\begin{aligned}\tilde{P}(x,y := \mathbb{P}(\tilde{X}_{n+1} = y \mid \tilde{X}_n = x) &= \mathbb{P}(X_{N-(n+1)} = y \mid X_{N-n} = x) \\ &= \frac{\mathbb{P}(X_{N-(n+1)} = y, X_{N-n} = x)}{\mathbb{P}(X_{N-n} = x)} \overset{\text{if reversible}}{=} \frac{\pi(y)P(y,x)}{\pi(x)}\end{aligned}$$

The forward and backward Markov chain have the same transition Matrix if and only if it is reversible.

Sample paths of Brownian motion



Six independent paths

Each curve is a realization of standard Brownian motion. All paths begin at zero and evolve independently.

$$B_0^{(k)} = 0, \quad k = 1, \ldots, 6.$$

The animation loops as each trajectory unfolds.

The Idea of Diffusion

Start with a weighted semigroup average

Standard diffusion

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t$$

The exponential kernel gives the resolvent

Start with a weighted semigroup average

For a kernel ϕ, define

$$P_\phi f(x) := \int_0^\infty \phi(t)\, P_t f(x)\, dt.$$

Taking the exponential kernel

$$\phi_p(t) := e^{-pt}, \qquad p > 0,$$

recovers the resolvent, which is the Laplace transform of the semigroup:

$$R_p \equiv P_{\phi_p}.$$

Build a class from exponential kernels

Consider finite linear combinations of exponentials:

$$\mathcal{C} := \left\{ \phi : [0,\infty) \to \mathbb{R} \;\middle|\; \phi(t) = \sum_{i=1}^{n} c_i e^{-p_i t},\ c_i \in \mathbb{R},\ p_i > 0,\ n \in \mathbb{N} \right\}.$$

By linearity, every $\phi \in \mathcal{C}$ satisfies

$$\langle P_\phi f, g \rangle_\nu = \sum_{i=1}^{n} c_i \langle R_{p_i} f, g \rangle_\nu = \langle f, \widehat{P}_\phi g \rangle_\nu.$$

Linearity carries the resolvent adjoint relation to finite exponential sums.

Previous result For each $p > 0$, $R_p = P_{\phi_p}$.

Density extends the identity to C_0

The exponential class is dense in $C_0([0,\infty))$. Thus, for each $\phi \in C_0([0,\infty))$, there is a sequence $\phi_n \in \mathcal{C}$ such that

$$\lim_{n\to\infty} \sup_{t\geq 0} \left|\phi_n(t) - \phi(t)\right| = 0.$$

Uniform approximation $\Longrightarrow$ extension beyond exponentials